Google Cloud

Stages of the data lifecycle

Hello, and welcome to this exploration of the data lifecycle!

Understanding how data moves through its entire life will help you, as a cloud professional, stay organized and keep your work as efficient as possible.

Let’s begin with an example.

Maybe your analytics team is assigned a new project.

It can be tempting for everyone to want to get right to work.

But organizing project phases and creating a plan before you begin helps you get much better results.

Likewise, knowing who’s responsible for which tasks creates a streamlined project!

Any data project will benefit from understanding the data lifecycle.

This is the sequence of stages that data experiences, which include: plan, capture, manage, analyze, archive, and destroy.

Let’s dive into each one a bit further.

As its name suggests, the plan stage begins before you start any analysis.

Usually it involves answering a particular business question or meeting an objective.

It's important that all project outcomes are focused on this key point.

A typical business question is “How can we increase user engagement with our product?.”

And an objective might be “We will reduce our plastic use by 50 percent by using only recycled materials in our package design.”

Next, team members decide what data types to collect, which data management processes to follow, and who is responsible for each stage.

During the planning stage, the ideal project outcomes and how to measure success are defined.

Next is the capture stage, when data is collected from different sources.

This may include outside resources, like publicly available datasets, or it may be data that's generated from within the organization.

This stage is also when gaps are identified in an organization's current data collection and are improved through iteration.

For example, maybe business leaders want to know how many users have downloaded their app.

But when reviewing the existing data collection plan, an analyst notes that they only capture data

about how many users have created an account, but not always how many app downloads have occurred.

The analyst may need to work with the engineering team to figure out if it's possible to get this data logged.

Then, it's time to process the captured data.

This is important because raw data directly from the source is usually not in a usable format for analytics.

So, a data engineer needs to clean the data and transform the data.

They may also compress the data and encrypt the data for security purposes.

Now, we've come to the manage stage.

The goal of this stage is to ensure proper data maintenance.

This includes safe and secure data storage.

Keep in mind that the manage stage is an ongoing process throughout the project.

In other words, the raw data collected in the capture stage, the transformed data from the process

stage, and any business logic that analysts establish all must be managed securely on an ongoing basis.

Next, is the analyze stage.

This is when analysts use data to answer the business question or help meet the business objective generated during planning.

At this point, team members review the compiled data to find trends, create visualizations, and suggest business recommendations based on data insights.

Then, in the archive stage, the data engineer may store data for later use, if needed.

Finally, when data is no longer useful to the organization, it's destroyed.

Although it might seem like a waste to destroy data, this is an important stage to ensure

that sensitive data cannot be stolen, to support privacy protection guidelines, and meet other compliance requirements and regulations.

For example, this may keep data practices in compliance with GDPR, the General Data Protection Regulation, in the European Union.

This regulation covers the data that an organization is allowed to hold.

Ok, one final point about the data lifecycle: None of these stages work without data team members.

You just learned that data analysts primarily work during the analysis stage.

They collaborate with data architects, who design the plan for managing the data.

Data engineers build the infrastructure for the data.

And data scientists use the data to create models to understand the data.

With these talented people in place, your organization will have a point person for each stage of the data lifecycle.

And everyone will have a clear idea of where projects and data are headed!